

## **Data Science (Python) - 45 Marks**

1. Matrix Transposition: - 2 Marks
  - Question: Write a function to transpose a given matrix. The matrix is represented as a list of lists.
  - Input: `[[1, 2, 3], [4, 5, 6], [7, 8, 9]]`
  - Output: `[[1, 4, 7], [2, 5, 8], [3, 6, 9]]`
2. Palindrome Check with Punctuation: - 3 Marks
  - Question: Write a function that checks if a given string is a palindrome, ignoring spaces, punctuation, and case.
  - Input: "A man, a plan, a canal: Panama"
  - Output: True
3. \* Anagram Detection: - 5 Marks
  - Question: Write a function that checks if two given strings are anagrams of each other.
  - Input: "listen", "silent"
  - Output: True
4. \* Generate Pascal's Triangle: - 5 Marks
  - Question: Write a function that generates the first n rows of Pascal's triangle.
  - Input: 5
  - Output: `[[1], [1, 1], [1, 2, 1], [1, 3, 3, 1], [1, 4, 6, 4, 1]]`
5. Longest Common Subsequence: - 2 Marks
  - Question: Write a function to find the longest common subsequence of two given strings.
  - Input: "abcde", "ace"
  - Output: "ace"
6. Flatten Nested List: - 2 Marks
  - Question: Write a function to flatten a nested list structure.
  - Input: `[1, [2, [3, 4], 5], 6]`
  - Output: `[1, 2, 3, 4, 5, 6]`
7. \* Binary Tree Inorder Traversal: - 5 marks
  - Question: Write a function to perform an inorder traversal on a given binary tree.
  - Input: A binary tree with root node having value 1, left child 2, and right child 3
  - Output: `[2, 1, 3]`
8. Finding the Kth Largest Element: - 2 Marks
  - Question: Write a function to find the Kth largest element in an unsorted list.
  - Input: `[3, 2, 1, 5, 6, 4]`, `k = 2`
  - Output: 5
9. \* Find Subarray with Given Sum: - 5 Marks
  - Question: Write a function to find a contiguous subarray within a given array that has a specified sum.
  - Input: `[1, 2, 3, 7, 5]`, `sum = 12`
  - Output: `[2, 3, 7]`

10. Calculate Expression Value: - 1 Marks

- Question: Write a function to evaluate a mathematical expression given as a string, handling basic operators (+, -, \*, /) and parentheses.
- Input: "3 + 5 \* 2 - (6 / 3)"
- Output: 10.0

11. \* String Compression: - 5 Marks

- Question: Write a function that compresses a string such that 'aaabb' becomes 'a3b2'. If the compressed string is not shorter, return the original string.
- Input: "aaabb"
- Output: "a3b2"

12. \* Rotate Matrix: - 5 Marks

- Question: Write a function to rotate a given n x n matrix by 90 degrees clockwise.
- Input: [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
- Output: [[7, 4, 1], [8, 5, 2], [9, 6, 3]]

13. \* Implement Queue Using Stacks: - 5 Marks

- Question: Write a class to implement a queue using two stacks.
- Input: enqueue(1), enqueue(2), dequeue()
- Output: 1

14. \* Balanced Parentheses: - 5 Marks

- Question: Write a function to check if a string containing parentheses, curly braces, and square brackets is balanced.
- Input: "{[()]}"
- Output: True

15. \* Convert Integer to Roman Numeral: - 5 Marks

- Question: Write a function to convert an integer to a Roman numeral.
- Input: 58
- Output: "LVIII"

16. \* Word Ladder: - 5 Marks

- Question: Write a function that finds the length of the shortest transformation sequence from a start word to an end word, given a dictionary of valid intermediate words.
- Input: start = "hit", end = "cog", dict = ["hot", "dot", "dog", "lot", "log", "cog"]
- Output: 5

17. Find Peak Element: - 3 Marks

- Question: Write a function to find a peak element in an array. An element is a peak if it is greater than its neighbors.
- Input: [1, 2, 3, 1, 6, 7]
- Output: 3